

Project Specification - Simple Texture Mapping

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1 Background

A core aspect in real time computer graphics is visual fidelity. While it would be possible to add detail through more vertices and more complex geometry that is computationally expensive. A much cheaper and simpler way to improve the visual fidelity of a renderer is to add textures to the geometry. This, of course, requires those textures to be properly mapped to the rendered geometry.

2 Problem

We already have a rasterizer from the laboratory work in the rendering track. But it is very simplistic with only uniform colours on each surface, making it look quite dull. The goal of this project is to add perspective correct texture mapping to the rasterizer in order to improve its visual fidelity, though making good looking textures is beyond the scope this project.

3 Implementation

3.1 Image Loading

Writing a parser for image files is not within the scope of this project, as such we will be using the stb_image library created by Sean Barrett [1] in order to easily load arbitrary image formats, which would ideally speed up the development process.

3.2 Texture Mapping

The implementation for this will be based on the methods described by Paul Heckbert [2]. More specifically we will be focusing on the incremental techniques described. The reason for this is that the rasterizer is already rendering incrementally so those techniques should map pretty cleanly to the existing implementation. It will also be important to make sure the texture are perspective correct rather than being skewed or otherwise looking weird.

References

- [1] Sean Barrett. *stb*. <https://github.com/nothings/stb>. Accessed: 2026-05-12. 2014.
- [2] Paul Heckbert. “Fundamentals of Texture Mapping and Image Warping”. In: (May 1998).